

Low-Complexity FPGA-Based PR-9QAM Detection for Photonics-Aided 306-GHz FTN Wireless System

Yikai Wang^(1,2), Min Zhu^{(1,2)*}, Yuancheng Cai⁽²⁾, Long Zhang⁽²⁾, Junjie Ding⁽²⁾, Jiao Zhang⁽²⁾, Bingchang Hua⁽²⁾, Mingzheng Lei⁽²⁾, Zhigang Xin^(1,2), Yiting Yan^(1,2), Jianwei Chen^(1,2), Jiankang Li^(1,2), Xiongwei Yang⁽³⁾, Kaihui Wang⁽⁴⁾, and Jianjun Yu^(2,4)

⁽¹⁾ Southeast University, Nanjing 210096, China

⁽²⁾ Purple Mountain Laboratories, Nanjing, 211111, China

⁽³⁾ Xi'an University of Posts and Telecommunications, Xian, 710121, China

⁽⁴⁾ Fudan University, Shanghai, 200433, China

* Author e-mail address: minzhu@seu.edu.cn

Abstract *This paper employs PR coding for FTN transmission, yielding a PR-9QAM signal and motivating low-complexity CMMA and carrier recovery design. The algorithms are implemented on an FPGA platform and validated in a photonics-aided 306 GHz link with error-free 30 Gbps over 3 m.*

Introduction

Terahertz (THz) communications, enabled by extremely abundant spectral resources, have emerged as a key technology for future networks requiring ultra-high capacity and ultra-low latency. [1-5]. However, system capacity enhancement is often limited by the analog bandwidth and sampling rate of practical devices. In particular, in FPGA-based real-time signal processing platforms, the finite sampling rate of analog-to-digital converters (ADCs) leads to severe spectral aliasing when the signal bandwidth exceeds the Nyquist limit, thereby restricting the achievable symbol rate.

To mitigate this bandwidth limitation, existing approaches typically rely on optoelectronic joint down-conversion to shift the signal spectrum to baseband [6–8]. However, such hardware-intensive solutions inevitably introduce additional link losses and significantly increase system complexity and cost. In contrast, partial response (PR) coding in the digital domain has emerged as a more cost-effective solution, enabling faster-than-Nyquist (FTN) transmission by compressing the signal spectrum [9]. Nevertheless, PR coding transforms constant-modulus QPSK symbols into a non-constant modulus PR-9QAM constellation, which places more stringent requirements on subsequent equalization and carrier recovery, thereby increasing overall DSP complexity.

To address bandwidth constraints, we demonstrate a real-time FTN coherent THz system employing PR coding. Over a 3 m wireless link at 306 GHz, a 30 Gbps signal is successfully processed using real-time FPGA-based DSP. We introduce a low-complexity cascaded multi-modulus algorithm (CMMA), utilizing an Alpha-Max Beta-Min [10] approximation for modulus computation, together with a magnitude-threshold-based carrier recovery scheme for efficient PR-9QAM demodulation. Experimental results show error-

free transmission at a bandwidth retention ratio (BRR) of 0.85. Stable real-time FPGA processing is maintained under strong spectral compression, confirming the effectiveness of the proposed low-complexity DSP for high spectral efficiency FTN transmission, enabling a hardware-efficient solution for THz mobile fronthaul applications.

Principles

To achieve FTN transmission, this work employs the PR encoding method [9] for spectral compression. The encoding process is formulated in Eqs. (1) and (2), where M_{th} represents the modulation order:

$$b_n = \text{mod}\left(\frac{X_n + M_{th} - 1}{2} - b_{n-1}, M_{th}\right) \quad (1)$$

$$B_n = 2 \cdot (b_n + b_{n-1} - M_{th} + 1) \quad (2)$$

By introducing controlled Inter-Symbol Interference (ISI) and utilizing a brick-wall filter (BWF), PR encoding effectively circumvents the Nyquist bandwidth limit, significantly enhancing the spectral efficiency of the system. The decoding process at the receiver is performed according to :

$$X_n = \text{mod}\left(\frac{B_n}{2} + M_{th} - 1, M_{th}\right) \cdot 2 - M_{th} + 1, n \geq 1 \quad (3)$$

Post PR-encoding, the original constant-modulus QPSK constellation expands into a PR-9QAM structure with multiple amplitude levels. This constellation exhibits distinct non-constant modulus characteristics and a non-uniform probability distribution. The traditional constant modulus algorithm (CMA) suffers from convergence degradation due to the presence of zero-amplitude components. To address this, a CMMA is implemented to adapt to the radial distribution of the signal. The modulus error $\varepsilon(i)$ and the composite error signal $e(i)$ used for filter weight updates are defined as follows:

$$\varepsilon(i) = \left| |z(i)| - R_{ref1} | - R_{ref2} | - R_{ref3} \right| \quad (4)$$

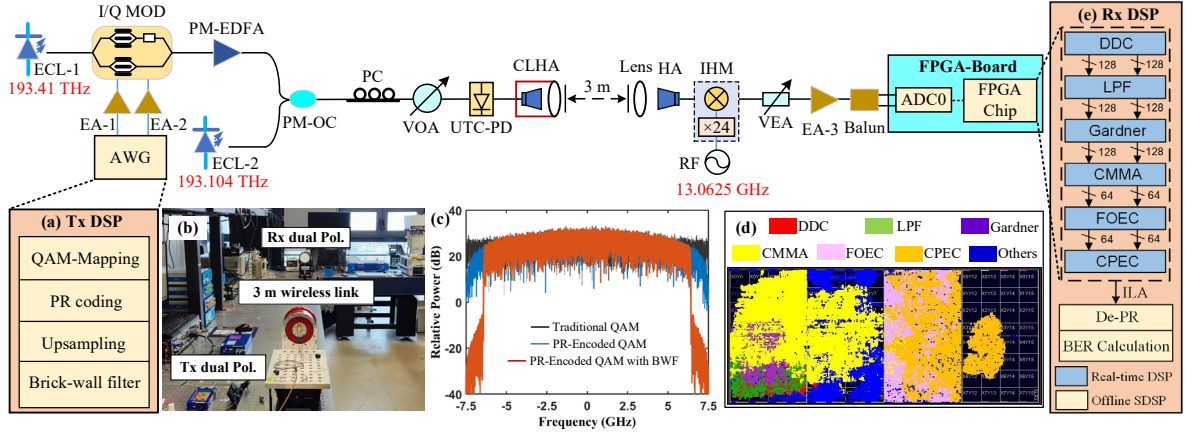


Fig. 1: Experimental setup of the FPGA-enabled photonics-aided THz wireless transmission system: (a) Transmitter DSP block diagram; (b) Photograph of the 3-meter wireless transmission link; (c) Comparison of Signal Spectra; (d) FPGA placement and routing; (e) Receiver DSP block diagram.

$$e(i) = \text{sign}(\left| |z(i)| - R_{ref1} | - R_{ref2} | - R_{ref3} \right|) \cdot \text{sign}(\left| |z(i)| - R_{ref1} | - R_{ref2} \right|) \cdot \text{sign}(\left| |z(i)| - R_{ref1} \right|) \quad (5)$$

where R_{ref1} , R_{ref2} and R_{ref3} represent the reference statistical radii for the PR-9QAM rings. To reduce hardware implementation complexity, the Alpha-Max Beta-Min approximation is employed to simplify the complex magnitude operation $|z(i)|$:

$$|z(i)| \approx \max(|z_r(i)|, |z_q(i)|) + 0.5 \cdot \min(|z_r(i)|, |z_q(i)|) \quad (6)$$

Substituting Equation (6) into the equalization process significantly reduces multiplications and square-root operations in the real-time digital signal processing (DSP) subsystem.

The PR-9QAM constellation exhibits inconsistent phase rotation characteristics across its radial layers, which causes conventional carrier recovery algorithms to converge slowly under non-coherent superposition. This paper proposes a classified estimation strategy based on amplitude thresholding. By leveraging the fact that the second-ring constellation points account for 50% of the distribution and possess strict fourth-order rotational symmetry, we selectively extract this layer for frequency and phase offset estimation. This approach effectively suppresses interference from origin noise and phase ambiguity in the outer ring. Simulation results indicate that the proposed strategy achieves reliable synchronization performance while significantly reducing the required observation length for parameter estimation.

Experimental Setup

This study establishes the FPGA-enabled photonics-aided THz wireless transmission experimental platform shown in Fig. 1. At the transmitter, the random binary sequence is mapped into QPSK symbols, followed by PR encoding and 2-fold up-sampling. A BWF is then

applied for spectral compression, as illustrated in Fig. 1(c). The resulting baseband signal is loaded into an arbitrary waveform generator (AWG) to generate a 15 GBaud electrical signal, which is amplified by an electrical amplifier (EA) to drive an I/Q modulator. This modulates the optical carrier from an external cavity laser (ECL-1). The modulated signal is then boosted by a polarization-maintaining erbium-doped fiber amplifier (PM-EDFA) and combined with a 306 GHz offset local oscillator from ECL-2 via a polarization-maintaining optical coupler (PM-OC). After optimizing the polarization and power via a polarization controller (PC) and a variable optical attenuator (VOA), the signal is converted by a uni-traveling-carrier photodiode (UTC-PD) into a 306 GHz THz wave, subsequently radiated via a high-gain cylindrical lens horn antenna (CLHA).

At the receiver, the THz signal is focused by a 10 cm lens and captured by a 26 dBi horn antenna (HA), followed by down-conversion to a 7.5 GHz intermediate frequency (IF) signal using an integrated harmonic mixer (IHM). The IF signal is subsequently adjusted by a variable electrical attenuator (VEA) and amplified by a 35 dB EA-3. The amplified single-ended signal is then converted into a differential signal via a balun to match the input requirements of the high-speed ADC. The signal is finally digitized by a real-time processing platform based on a XCVU13P FPGA, equipped with a 30 GSa/s, 8-bit ADC.

The receiver-side DSP flow is shown in Fig. 1(e), with all modules operating under a unified 230.4 MHz clock. First, digital down-conversion (DDC) processes the 128-channel parallel ADC data, generating 128 parallel streams of in-phase and quadrature components. A low-pass filter (LPF) is then applied to suppress image components and out-of-band noise, recovering the baseband signal. Gardner timing recovery is subsequently employed for clock skew correction

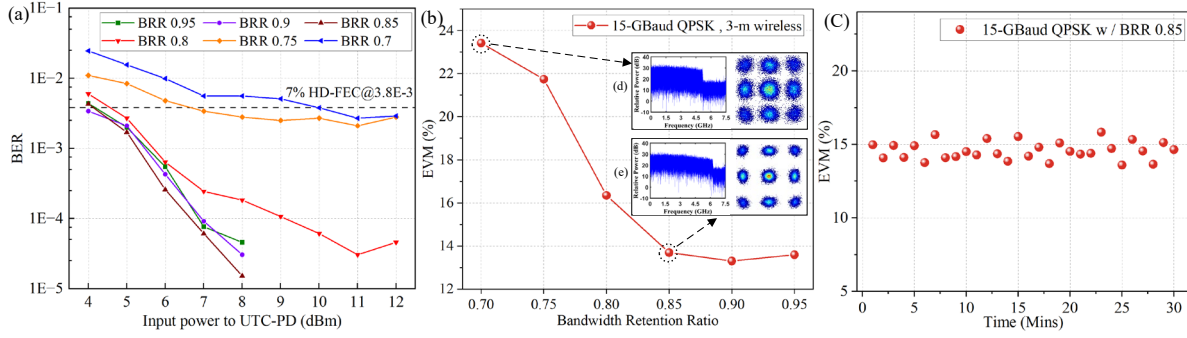


Fig. 2: Real-time experimental characterization of the PR-FTN system: (a) BER versus ROP under different BRRs; (b) EVM versus BRR at 11-dBm ROP; (c) Long-term stability test of the system EVM.

(proportional coefficient 2^{-3} and integral coefficient 2^{-10}), while maintaining the 128-way parallel structure. The corrected signal is then fed into a 13-tap CMMA equalizer with a T/2 structure. The module adopts a highly parallel pipelined architecture, enabling linear impairment compensation and factor-of-two down-sampling while ensuring real-time high-throughput processing.

Next, both the frequency offset estimation and compensation (FOEC) module and the carrier phase estimation and compensation (CPEC) module employ an amplitude-threshold-based selection strategy, extracting only the second-order constellation points with strong coherent gain. FOEC is implemented using a fourth-power FFT algorithm with a 128-point processing window and a 65,536-point update interval. CPEC performs phase tracking using a 64-point Viterbi-Viterbi algorithm. The compensated real-time data are captured by an integrated logic analyzer (ILA) for offline De-PR decoding and bit-error-rate (BER) calculation. The FPGA placement and routing results are shown in Fig. 1(d).

Experiment results and discussion

We first evaluate the impact of the BRR, defined as the ratio of the BWF cutoff bandwidth to the original signal bandwidth, on system BER performance. Fig. 2(a) shows the BER as a function of the received optical power for different BRR values. The results indicate that when $\text{BRR} \geq 0.85$, error-free transmission is achieved at 8 dBm, demonstrating the strong compensation capability of the proposed scheme under stringent bandwidth constraints. When BRR decreases to 0.8, severe ISI exceeds the equalizer limit, resulting in an error floor. For BRR values of 0.75 and 0.7, performance degrades significantly, reaching the 7% hard-decision forward error correction (HD-FEC) threshold of 3.8×10^{-3} at received powers of -7 dBm and -10 dBm, respectively. These results confirm that excessive bandwidth compression induces pronounced ISI, which fundamentally limits further spectral reduction.

Fig. 2(b) illustrates the EVM versus BRR at 11

dBm, where EVM decreases with increasing BRR and reaches a minimum at $\text{BRR} = 0.9$. The insets in Fig. 2(b), namely Figs. 2(d) and 2(e), present the constellations and spectra for $\text{BRR} = 0.7$ and 0.85, respectively, providing intuitive insight into the trade-off between spectral efficiency and ISI mitigation. Considering both capacity and link margin, $\text{BRR} = 0.85$ is selected as the optimal operating point. The long-term stability measurement shown in Fig. 2(c) confirms negligible EVM fluctuation over time, validating the robustness of the proposed system.

Table 1 summarizes the DSP resource utilization of the receiver implemented on a XCVU13P FPGA. Although the core modules, including CMMA, FOEC, and CPEC, dominate the logic consumption, the overall resource usage remains low. This low-complexity hardware implementation satisfies the low-power requirements of optical transmission subsystems and provides a viable solution for future high-speed, high-capacity optical networks.

Tab.1: Comparison of DSP resource overhead on FPGA.

Module /Type	DSP48E2	LUTs	FF
DDC	0(0%)	2014(0.1%)	5683(0.2%)
LPF	0(0%)	28160(1.6%)	30720(0.9%)
Gardner	638(5.2%)	36048(2.1%)	31239(0.9%)
CMMA	3120(25.4%)	201535(11.7%)	289431(8.4%)
FOEC	1800(14.7%)	175157(10.1%)	220301(6.4%)
CPEC	576(4.7%)	88457(5.1%)	104736(3.0%)

Conclusion

This work experimentally demonstrates a real-time FTN coherent THz system using PR coding. By implementing a full FPGA-based receiver with optimized CMMA and carrier recovery, error-free 30 Gbps transmission is achieved at 306 GHz. This hardware-efficient architecture significantly reduces computational complexity, providing a vital reference for high-capacity THz mobile fronthaul.

Acknowledgements

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References

- [1] Dat, P.T. and Akahane, K., 2025. Photonics-Enabled Millimeter-/Terahertz-Wave Signal Processing and Applications. *Journal of Lightwave Technology*. DOI: 10.1109/JLT.2025.3569707
- [2] Yang, X., Yu, J., Zhao, X., Wang, M., Wei, Y., Wang, C., Zhang, L., Long, J., Zhou, W., Wang, K. and Zhao, F., 2025, September. High-Laser Linewidth-Tolerance Photonics-aided 300 GHz Terahertz Wireless Transmission System. In 2025 European Conference on Optical Communications (ECOC) (pp. 1-4). IEEE. DOI: 10.1109/ECOC66593.2025.11263007
- [3] Long, J., Wang, C., Yu, J., Zhang, L., Sang, B., Wu, Y., Yang, X., Wei, Y., Wang, K., Zhou, W. and Zhang, J., 2024, September. Demonstration of 562.5-Gbps 2× 2 MIMO terahertz-wave signal transmission at 322 GHz with SFO compensation. In ECOC 2024; 50th European Conference on Optical Communication (pp. 551-554). VDE. <https://ieeexplore.ieee.org/abstract/document/10926378>
- [4] Zhu, M., Tong, W., Ding, J., Zhang, J., Hua, B., Cai, Y., Lei, M., Huang, Y., Yu, J. and You, X., 2025. Large-capacity long-distance photonics-aided terahertz wireless communication system: key techniques and experimental demonstration. *Science China Information Sciences*, 68(12), p.220301. <https://doi.org/10.1007/s11432-025-4604-4>
- [5] Harter, T., Fullner, C., Kemal, J.N., Ummethala, S., Steinmann, J.L., Brosi, M., Hesler, J.L., Bründermann, E., Freude, W., Randel, S. and Koos, C., 2019, September. Generalized Kramers-Kronig receiver for 16QAM wireless THz transmission at 110 Gbit/s. In 45th European Conference on Optical Communication (ECOC 2019). Stevenage UK: IET. DOI: 10.1049/cp.2019.1000
- [6] Meng, J., Ding, J., Wang, Y., Zhang, L., Zhang, J., Cai, Y., Hua, B., Lei, M., Yu, J. and Zhu, M., 2025, June. FPGA-Based 30-Gbit/S Fiber-THz Wireless Seamless Integrated Communication System. In 2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC) (pp. 1-4). IEEE. DOI: 10.23919/OECC/PSC62146.2025.11111003
- [7] Wang, C., Yu, J., Li, X., Gou, P. and Zhou, W., 2018. Fiber-THz-fiber link for THz signal transmission. *IEEE Photonics Journal*, 10(2), pp.1-6. 10.1109/JPHOT.2018.2809433
- [8] Zhu, M., Zhang, J., Hua, B., Lei, M., Cai, Y., Tian, L., Wang, D., Xu, W., Zhang, C., Huang, Y. and Yu, J., 2023. Ultra-wideband fiber-THz-fiber seamless integration communication system toward 6G: architecture, key techniques, and testbed implementation. *Science China Information Sciences*, 66(1), p.113301. <https://doi.org/10.1007/s11432-022-3565-3>
- [9] Yang, X., Yu, J., Zhang, L., Wei, Y., Tan, J., Zhao, F., Zhou, W. and Wang, K., 2025. High spectral efficiency 65536 QAM mobile fronthaul based on filtered delta-sigma modulation scheme. *Journal of Lightwave Technology*. DOI: 10.1109/JLT.2025.3541039
- [10] Assim A.A. ASIC implementation of high-speed vector magnitude & arctangent approximator. *Computing, Telecommunications and Control*, 2021, Vol. 14, No. 4, Pp. 7–14. DOI: 10.18721/JCSTCS.14401